PN. L003094143NCP-AA

Potential Failure Mode and Effects Analysis

No. : TRN023-1-1

Size : M6-1.0 X 12

Description : (T30) 6 LOBE PAN HEAD MACHINE SCREW, CLASS 8.8 10B21

Process Responsibility :

Model Years(s)/Vehicle(s) :

Key Date : November 28, 2024

FMEA Date (Orig.) : November 28, 2024

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Process Function Requirement 製程功能要求	Potential Failure Mode 潛在的失敗模式	Potential Effect(s) of Failure 失敗可能造成原因/機制	SEV（S）	CLASS	Potentail Cause(s) / Mechanism(s) of failure 失敗的潛在原因/機制	OCCUR（O）	Current Process Controls Prevention 預防性控制	Current Process Controls Detection 偵測性控制	DETEC（D）	RPN	Recommended Action 建議採取之行動	Responsibility & Target Completion DateResponsibility & Target Completion Date	Action Result 行動的結果				
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1 Wire Material	1-1 Get rust	Product get rust	5		Surface treatment failure	2	Staff education and training	Color board check	6	60							
	1-2 Wrong dimension	Not working	6		Tooling broke	1	Measurement instrument periodic verification	Dimension check	5	30							
	1-3 Wrong material	Material cannot meet request mechanical property	6		Production failure Wrong label	2	Check material certificate	Incoming wire inspection	5	60							
	1-4 Material is too soft	Material decarburization	5		Production procedure is incorrect	1	Measurement instrument periodic verification	Metallographic examination	5	25							

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2 Heading Forming Process.Heading Machine Setup.Heading Tooling	2-1 Head diameter is out of tolerance.	*Unable to screw tight	6		*Material feed is too long *Material feed shorts *Feeding wheel's parameter setup is incorrect *Wire diameter is not even	2	Weekly, 2nd and 3rd level machine maintenance *Check wire and separate NG part with RED label and storage at different location. *Incoming wire diameter should be the same. If there is difference out of tolerance, personnel should mark in the label.	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
	2-2 Material feed setting failed	*Head diameter is unstable. *Process is unstable	6		*The adjustment of material feed is not appropriate *The cutting hole is too small	2	*Adjust the length of material feed *Wire feed length adjustment	*Measurement. *First piece inspection *Self inspection *In-process inspection	5	60							
	2-3 The head height is out of tolerance.	* Too thick: cannot be assembled. * Too thin: easy to distort and attrition	6	K	*The adjustment of material feed is not appropriate *Poor machine precision *The punch-pin and tooling is adjusted incorrectly	2	*Weekly, 2nd and 3rd level machine maintenance *Do self inspection when punch-pin is changed *Adjust second stage tooling per punch-pin dimension *Tooling changed	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *On-line Statistical Process Control *Last piece inspection	5	60							

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	2-4 Heading Machine Setup-Head height adjustment fail	*Head height is unstable. *Process is unstable	6		*The adjustment of material feed is not appropriate *Poor machine precision *The punch-pin and tooling is adjusted incorrectly	2	*Adjust second stage tooling per punch-pin dimension *Tooling changed	*Measurement. *First piece inspection *Self inspection *In-process inspection	5	60							
	2-5 B:Radius of the raised portion of a head is out of tolerance.	*Poor appearance	6		*Tooling's radius is out of tolerance.	2	*Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-6 Underhead radius is out of tolerance	*Too big: unable to assemble; too small: easy to break	6		* Tooling radius design fail *Wrong tooling dimension	2	*Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed *In-process inspection twice per day *Last piece inspection	5	60							
	2-7 Transition diameter is out of tolerance.	*Too big: unable to assemble; too small: easy to break	6		*Tooling radius design fail *Wrong tooling dimension	2	*Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-8 Head concentricity	*Easy to strip from assembly	6		*Poor tooling concentricity	2	*Weekly, 2nd and 3rd level machine maintenance *Return NG tooling to the vendor to amend *Adjust 1st and 2nd blow tooling	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							

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	2-9 Wrong T30 recess	*Wrong recess can't be used	6		*Choose wrong punch-pin *Punch pin damaged	2	*Punch-pin incoming inspection *Check recess with P guage when dimension or punch-pin is changed	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection- 1.Change punch-pin 2.every 2/3 plate *In-process inspection twice per day *Last piece inspection	6	72							
	2-10 T30's penetration depth is out of tolerance.	*Too shallow: bit is easy to jump out during assembly. Too deep: the head broken is easy to break.	6		*Punch-pin is damage or/and overuse	2	*Punch-pin incoming inspection *Limit punch-pin's usage life	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection- 1.Change punch-pin 2.Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
	2-11 Recess dimension is wrong, and the NO GO gauges enter to the penetration depth.	*Too deep: easy to slip from assembly	6		*Recess dimension is wrong, and the NO GO gauges enter to the penetration depth.	2	*Check penetration depth when dimension is changed or adjusted	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection- 1.Change punch-pin 2.Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							

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	2-12 Recess Diameter	*Too Big: bit is easy to slip during assembly *Too Small: bit is easy to jump out of recess during assembly	6		*Wrong punch-pin design *Wrong manufacturing parameters	2	*When the specification is changed or adjusted with P guage GO & NOT GO. *Measure recess diameter with projector *Check depth again with P gauge	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-13 Recess concentricity	*Easy to strip from assembly	6		*Poor punch-pin accuracy	2	*Request punch-pin's concentricity *Weekly, 2nd and 3rd level machine maintenance	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-14 Perpendicularity	*Poor contact to the application's surface, and easy to loosen	6		*Poor tooling perpendicularity *Poor machine precision	2	*Return NG tooling to the vendor to amend *Weekly, 2nd and 3rd level machine maintenance *Adjust 1st and 2nd blow tooling	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-15 Wrong straightness	*Not easy to screw tight	6		*Out of blank diameter tolerance *Wrong thread die dimension *Wrong manufacturing parameters setup	2	*First piece inspection-Straightness *Go production after initial sample is OK	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							

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	2-16 The taper length is out of tolerance.	*Poor index function	6		*Wrong taper length	2	*Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-17 Wrong taper angle	*Poor index function	6		*Wrong taper angle of tooling	2	*Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-18 The taper diameter is out of tolerance.	*Wrong taper point diameter which is not easy to screw into the application.	6		*Wrong taper diameter of tooling	2	*Return NG tooling to the vendor to amend*	*First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-19 Underhead radius diameter too big	*Poor contact to application's surface	6		*Forming by 1st punch-pin is poor	2	*Weekly, 2nd and 3rd level machine maintenance *Re-adjust 1st punch-pin *Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.Tooling changed *In-process inspection twice per day *Last piece inspection	5	60							
	2-20 Wrong head marking	*Poor appearance and cannot be recognized	6		*Head forming failed at first stage *Wrong punch-pin marking	2	*Re-adjust manufacturing parameters *Re-design punch-pin	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							

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	2-21 The blank diameter is out of tolerance.	*Too big: Major diameter is too big too to screw into application. Too small: Major diameter will be too small to tighten into application.	6		*Wrong blank diameter	2	*Return NG tooling to the vendor and amend *Manufacture per work order's standard	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
	2-22 Blank length is out of tolerance	*Blank length is out of tolerance	6		*Tooling and punch-pin is too tight *Improper machine operation	2	*Confirm blank length after thread rolling process and record it into the system	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
2 Heading Forming Process.Heading Machine Setup.Heading Tooling	2-23 Heading Machine Setup-Improperly adjust the length	*Blank length is unstable *Process is unstable	6		*Wire feed length is unstable *Tooling and punch-pin is too tight	2	*Head forming length adjustment *Adjustment of heading setup	*Measurement. *First piece inspection *Self inspection *In-process inspection	5	60							

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	2-24 Appearance	*Poor apperance	6		*Head forming failed *Head crack	2	*Once head crack is found, return the wire 1.Mark with red lable immediately 2.Move to NG parts storage area and wait for QA's confirmation and treatment 3.Immediately notify QA department	*First piece inspection- 1.In First sample 2.Tooling changed *Self inspection Self-inspection every 2/3 plate *Increase the number of samples for head cracks *In-process inspection twice per day *Sample appearance comparance *Last piece inspection	5	60							
	2-25 Incorrect machine operation	*Process is unstable *Wrong dimension	6		*Personal factor *Machine not stable	2	*Measurement.	*Measurement. *First piece inspection *Self inspection *In-process inspection	5	60							
2 Heading Forming Process.Heading Tooling	2-26 Heading tooling management failure	*Wrong screw dimension	6		*Wrong tooling design *Wrong tooling dimension	2	*Tooling incoming inspection *Design tooling per specification	*Tooling inspection *First piece inspection- 1.In First sample 2.Tooling changed	5	60							
	2-27 Forging power too large(RPM)	*Tooling is easy to break *Head forming failed	6		Forging power setup failed	2	Setup machine force	*First piece inspection	5	60							
	2-28 Prevent from picking wrong wire	Do not meet customer's requirement.	6		Wrong hardness	2	Pick wire via the ERP system with auto-matching steel and number.	*The ERP system matches conditions when picking the material everytime.	5	60							

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	2-29 Head forming machine's detection instrument for error proof.	*produce wrong dimension parts	6		*Improper operation	2	*Prevent material shortage and punch-pin broke and cause foreign parts.	*Take standard parts as measurement.	5	60							
3 Thread rolling Machine thread	3-1 The major diameter is out of tolerance.	*Major diameter is oversize, driving torque is too great to screw completely; Major Diameter is undersize to screw tightly	6	k	*Out of blank diameter tolerance *Wrong thread die dimension	2	*Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed *Statistical Process Control *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
	3-2 The minor diameter is out of tolerance.	*Too big- excess driving torque needed and not easy to screw in; too small- insufficient torque.	6		*The blank diameter is out of tolerance.	2	*Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed *In-process inspection twice per day *Last piece inspection	5	60							
	3-3 The pitch diameter is out of tolerance.	*Too big, the part need excess driving torque needed and is not easy to screw in. Too small, the part cannot screw tight to the application.	6		*The blank diameter is out of tolerance.	2	*Check thread die's dimension from initial inspection *Return NG thread die to the vendor to amend *Batch number control	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							

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	3-4 Thread die adjustment	*Setup fail to form correct major, minor diameter and pitch diameter	6		*Failed on thread die adjustment *	2	*Adjust material feed timing *Mirco adjust the distance between two thread rolling die	*Measurement. *First piece inspection	5	60							
	3-5 The thread angle is out of tolerance.	*Wrong design *Cannot assmeble	6		*Wrong thread angle of thread die	2	* *Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							
	3-6 Wrong pitch	*Wrong design *Cannot assmeble	6		*Wrong pitch of thread die *Poor machine precision	2	*Weekly, 2nd and 3rd level machine maintenance *Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							
	3-7 Wrong transition thread length	*Thread stripped and is unable to tight closer to application	6		*Wrong thread die design	2	*Start mass production when sample pass; amend thread die if NG	*First piece inspection- 1.In First sample 2.When thread die changed *In-process inspection twice per day *Last piece inspection	5	60							
	3-8 Wrong taper angle	*Poor index function	6		*Wrong tooling taper angle design result in wrong taper angle	2	*Wrong taper angle of tooling *Return NG tooling to the vendor and amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							

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	3-9 The taper length is out of tolerance.	*Too short: poor index function *Too long: Unable to screw tight	6		*Wrong tooling hole design results in wrong taper length	2	*Re-adjust setup after thread die is changed *Check tooling's dimension via initial sample inspection	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							
	3-10 Taper diameter is too big	*Not easy to screw into application *Result in wrong thread rolling taper diameter	6		*Wrong tooling taper diameter	2	*Confirm tooling's taper diameter after thread rolling process *Return NG tooling to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							
	3-11 Underhead radius is out of tolerance	*Too small: easy to break	6		*First thread is too close to underhead	2	*Readjust thread die	*First piece inspection- 1.In First sample 2.When thread die changed *In-process inspection twice per day *Last piece inspection	5	60							
	3-12 Wrong straightness	*Not easy to screw tight	6		*Out of blank diameter tolerance *Wrong thread die dimension *Wrong manufacturing parameters setup	2	*Check thread die's dimension via initial sample inspection *Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							
	3-13 Wrong perpendicularity	*Not easy to screw tight	6		*Wrong manufacturing parameters setup	2	*Check thread die's dimension via initial sample inspection *Return NG thread die to the vendor to amend	*First piece inspection- 1.In First sample 2.When thread die changed	5	60							

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	3-14 6g GO & NOT GO Ring gauge	*Thread stripped and is unable to tight closer to application	6		*The blank diameter is out of tolerance. *Thread die dimension cannot match manufacturing parameters setup	2	*Check thread die's dimension via initial sample inspectionGO & NOT GO Ring gauge *Adjust the tooling again or return it to the vendor to amend *Return NG thread die to the vendor to amend	*First piece inspection-1.In First sample 2.When thread die changed *Self inspection-Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	6	72							
	3-15 The length is out of tolerance.	*Too long: length is over the application and cannot use *Too short: unable to screw application tight	6		*Blank length is too long *Blank length is too short	2	*Adjust material feed timing *Mirco adjust the distance between two thread rolling die *Record correct blank length on the system as a precaution	*First piece inspection-1.In First sample 2.When thread die changed *Self inspection Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							
	3-16 Poor thread forming	*Thread forming failed *Unstable process result in poor appearance	6		*Wrong manufacturing parameters setup resulting in unstable process	2	*Adjust material feed timing *Mirco adjust the distance between two thread rolling die *Establish sample keeping plan for appearance check	*First piece inspection-1.In First sample 2.When thread die changed *Self inspection-Self-inspection every 2/3 plate *In-process inspection twice per day *Last piece inspection	5	60							

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		*Thread crack	6		*Thread die is worn without changing *Improper operation	2	*Thread die life management *Adjust material feed timing *Mirco adjust the distance between two thread rolling die	*First piece inspection- 1.In First sample 2.When thread die changed *In-process inspection once per day *Last piece inspection	6	72							
	3-17 Thread die managment failure	*Process is unstable *Wrong dimension *Incorrectly using thread die	6		*Wrong thread die design *Wrong thread die dimension	2	*Thread die incoming inspection *Thread die should be designed per specification	*Measurement. *First piece inspection- 1.In First sample 2.When thread die changed *Self inspection Self-inspection every 2/3 plate	5	60							
Thread rolling.Thread rolling die.Thread rolling machine setup	3-18 Damaged thread rolling die.AppearanceDeform, wrong dimension	*Process is unstable *Wrong dimension	6		*Personal factor *Machine not stable	2	*Adjust material feed timing *Mirco adjust the distance between two thread rolling die	*Measurement. *First piece inspection *Self inspection	5	60							
	3-19 Thread rolling die's parallelism	*Process is unstable *Wrong dimension	6		*Personal factor *Machine not stable *Thread rolling die is not parallel	2	*Correct square rule 90 degree's parralism *Both board need to be at the same height and with no burrs	*Measurement. *First piece inspection *Self inspection	5	60							
	3-20 RPM is too fast.	*Thread die is easily worn	6		Machine RPM setup failed	2	Setup machine speed	*First piece inspection	5	60							

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	3-21 Threading machine equips head-lifting device and pressure sensor for error proof	*Transition thread length is oversized and major dia. is wrong.	6		*Improper operation	2	*Prevent oversized grip gagging length and wrong major diameter.	*Take standard parts as measurement.	5	60							
4 Heat Treatment	4-1 Surface hardness exceeds or is insufficient	1.Easy to break 2.Thread damage cause stripping	6		Wrong carbon content	2	Measurement instrument periodic verification	(Surface hardness test)	5	60							
	4-2 Core hardness exceeds or is insufficient	1.Easy to break 2.Unable to afford strength	6		1.Tempering temperature is too low 2.Tempering time is insufficient 3.Tempering temperature is too high 4.Tempering time is too long	2	Measurement instrument periodic verification	Core hardness test	5	60							
	4-3 Torque test fail	1.Unable to afford strength or break	6		1.Tempering temperature is too low 2.Tempering time is insufficient 3.Tempering temperature is too high 4.Tempering time is too long	2	Measurement instrument periodic verification	Core hardness test	5	60							

Process Function Requirement 製程功能要求	Potential Failure Mode 潛在的失敗模式	Potential Effect(s) of Failure 失敗可能造成原因/機制	SEV（S）	CLASS	Potentail Cause(s) / Mechanism(s) of failure 失敗的潛在原因/機制	OCCUR（O）	Current Process Controls Prevention 預防性控制	Current Process Controls Detection 偵測性控制	DETEC（D）	RPN	Recommended Action 建議採取之行動	Responsibility & Target Completion DateResponsibility & Target Completion Date	Action Result 行動的結果				
													ACTION TAKEN 已採取行動	SEV（S）	OCCUR（O）	DETEC（D）	RPN
5 Coating	5-1 Insufficient coating thickness	Rust easily	6		1.Wrong coating process 2.Coating time is too short	2	Measurement instrument periodic verification	Coating thickness test	5	60							
	5-2 Salt spray testing failed	Rust easily	6		1.Wrong coating process 2.Coating time is too short	2	Measurement instrument periodic verification	Salt Spray Test	5	60							
	5-3 Appearance	Dull, rust Bubble	5		Insufficient coating thickness, and wrong coating time	2	Check the worksheet record	Profile projector inspection	5	50							
6 Optical sorting	6-1 Dimension, foreign parts	Too many defective parts	3		Machine setup fails	2	Measurement instrument periodic verification Compare and check	CCD sensor can auto-detect.	3	18							
7 Adhesive coating	7-1 Wrong coating length and angle	Poor self-locking and sealing function	6		Incorrect patching process	2	Measurement instrument periodic verification	Dimension check	5	60							
	7-2 Appearance	Wrong adhesive color	5		Wrong material compound ratio	2	Staff education and training	Color comparison check	5	50							
8 Final Inspection	8-1 Dimension NG not detected	Assemble is affected	6		Thread die setup failure Wrong inspection method	2	Calibrate measuring instruments regularly	Take and inspect samples from every batch	5	60							

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	Crack not detected	Assemble affected, function limited, customer dissatisfied	6		Wrong inspection method	2	Qualified final product inspector Standard sample establishment	Take and inspect samples from every batch	5	60							
	8-2 Unusual mechanical properties not detected	Assemble affected, function limited, customer dissatisfied	6		Thread die setup failure Wrong inspection method	2	Calibrate measuring instruments regularly	Take and inspect samples from every batch	5	60							
	8-3 Appearance	Assemble affected, function limited, customer dissatisfied	5		Wrong inspection method	2	Qualified final product inspector Standard sample establishment	Take and inspect samples from every batch	5	50							
9 Manual sorting	9-1 Foreign parts	Too many foreign parts	3		Improper machine adjustment	2	Establish standard sample for comparison	100% Inspection	6	36							
10 Packing	10-1 Unit weight error	The quantity is shortage or excess	3		Wrong calculation for the unit weight	2	Staff education and training Reconfirm packing worksheet	Packager need to double check and confirm the unit weight again before packing	6	36							
	10-2 Wrong Label	Label pieces does not match order's	3		Wrong packing unit input	2	Staff education and training Reconfirm packing worksheet	Packing leader have to double check the packing worksheet info	6	36							

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11 Shipping	11-1 Wrong Side mark	Does not match order information	3		Input error	2	Staff education and training Reconfirm packing worksheet	Packing leader have to double check the packing worksheet info	6	36							